🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.AT.9

Understand Inequalities and Their Solutions

Lesson 8-5 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can graph the solutions of an inequality on a number line.

Language Objective: I can explain my graph using the words number line, open circle, closed circle, and solution set.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Number line | **Open circle** | **Closed circle** | **Solution set** | **Inequality** | **Boundary point**

👆 *Tap any word to see its meaning and a picture.*

1. A line with numbers in order used to show values and solutions is a ________.

2. A hollow circle that shows a value is NOT included, used for $<$ or $>$, is an ________.

3. A filled-in circle that shows a value IS included, used for $\leq$ or $\geq$, is a ________.

4. All the values that make an inequality true form the ________.

5 A math sentence comparing two amounts with <, >, ≤, or ≥ is an

____________________.

6 The number where the solution set begins on the number line is the

____________________.

Write About the Math

Why did the detective use an open circle instead of a closed circle? What values of p are possible?

¿Por qué usó la detective un círculo abierto en vez de uno cerrado? ¿Qué valores de p son posibles?

Say your answer to a partner first. Then write it, using at least two of these words:

- [ ] **Number line** — *Recta numérica*
- [ ] **Open circle** — *Círculo abierto*
- [ ] **Closed circle** — *Círculo cerrado*
- [ ] **Solution set** — *Conjunto solución*
- [ ] **Inequality** — *Desigualdad*

Model: *Because 2:00 itself counts, I use a closed circle at 2.* — Students choose a closed circle at 2 (because 'no earlier than' includes 2) and shade to the right since later times satisfy $t \geq 2$.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective used an open circle because ____ means 12 is ____ included, so possible values are numbers ____ than 12.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Number line
2. **Notes 2:** Open circle
3. **Notes 3:** Closed circle
4. **Notes 4:** Solution set
5. **Notes 5:** Inequality
6. **Notes 6:** Boundary point
7. **Watch for this mistake:** *A common mistake in Graph Inequalities is treating ≥ and ≤ the same as > and < and drawing the wrong circle. For example, students often graph x ≥ 6 with an OPEN circle (as if it were x > 6), which wrongly excludes 6 even though 6 itself makes x ≥ 6 true. A second common mistake is choosing the right circle but shading the wrong direction — for x < 9 students sometimes shade right instead of left. Always check the symbol twice: once for the circle (does it have an 'or equal to' line?) and once for the shading direction (does the symbol point toward larger or smaller numbers?).*
8. **Try It 1:** A. Open circle at 15, shade right — *'Greater than' (>) is a strict symbol, so 15 is NOT included — open circle. Larger values work, so shade right.*
9. **Try It 2:** A. Closed circle at 4, shade left — *'Less than or equal to' (≤) includes 4 — closed circle. Smaller values work, so shade left.*